

Machine Learning-Based Performance Prediction and Analysis of UAV Propellers under Step-Throttle Conditions

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Abstract—Accurate performance prediction of unmanned aerial vehicle (UAV) propellers under dynamic operating conditions is crucial for propulsion system matching and flight efficiency optimization. Traditional static models often struggle to capture complex nonlinear aerodynamic behaviors. This study aims to significantly improve the prediction accuracy of propeller performance under various working conditions using a data-driven approach. To this end, a physical experimental test bench was established to conduct comprehensive step-throttle tests on two different models of propellers, systematically collecting real-world dynamic response data. Based on the experimental dataset, a machine learning-based prediction model was constructed. The model directly utilizes battery voltage and throttle command as input features to simultaneously and accurately predict key output metrics, including mechanical power, rotational speed, and overall efficiency. Experimental validation demonstrates that the proposed machine learning model exhibits excellent prediction accuracy and highly robust curve-fitting capabilities across different throttle steps. The results indicate that this data-driven framework effectively overcomes the limitations of conventional analytical methods, providing a highly reliable and efficient tool for evaluating and optimizing UAV propulsion systems.

Index Terms—machine learning, propeller performance, step-throttle test, performance prediction, data-driven model

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) and electric vertical take-off and landing (eVTOL) aircraft have experienced explosive growth in recent years, playing critical roles in fields ranging from aerial photography and agricultural mapping to logistics delivery and disaster inspection [1], [2]. The propulsion system is the core component of these flight platforms, directly determining crucial flight metrics such as maximum payload capacity, flight endurance, and maneuverability. Consequently, accurately evaluating and optimizing the energy efficiency of the propulsion system under various operating conditions has become a paramount research focus in aerospace engineering [3], [4].

In a typical multi-rotor propulsion system, the propeller serves as the primary aerodynamic component, converting the mechanical power of the motor into thrust. The dynamic matching between the battery, motor, and propeller is highly complex. During actual flight missions, UAVs frequently experience rapid changes in flight states, requiring continuous and dynamic adjustments of throttle commands. Simultaneously, the battery voltage naturally drops as the discharge process continues or fluctuates under sudden high-current loads. These dynamic operating conditions cause the propeller's performance metrics—specifically rotational speed, output power, and overall efficiency—to exhibit highly complex and nonlinear characteristics.

Traditionally, the performance evaluation of propellers relies heavily on static thrust stand tests and classical aerodynamic analytical models, such as Blade Element Momentum Theory (BEMT) [5], [6]. However, these conventional approaches present significant limitations in modern UAV design. First, traditional static testing typically focuses on steady-state points, which struggle to capture the transient aerodynamic responses and system delays inherent in dynamic operations, such as step-throttle conditions [7], [8]. Second, analytical models depend on numerous ideal assumptions and empirical coefficients. The highly nonlinear aerodynamic behaviors—including blade tip vortex, airflow separation, and the coupling effects of fluctuating voltage and variable rotational speed—are exceedingly difficult to describe accurately using simple analytical formulas [9]. As a result, there is a distinct gap between the theoretically calculated efficiency and the actual dynamic performance.

In recent years, the rapid advancement of artificial intelligence has provided new perspectives for solving complex engineering problems. Machine learning (ML) algorithms, with their exceptional capability to approximate complex nonlinear

mapping relationships, have shown great potential in modeling aerodynamic systems without requiring explicit physical equations [10]–[12]. By directly learning the intrinsic patterns from large-scale experimental data, ML models can bypass the complex aerodynamic derivations and provide highly accurate predictions [13]–[15].

To address the aforementioned pain points, this paper proposes an end-to-end performance prediction model for UAV propellers by combining rigorous physical bench testing with advanced machine learning algorithms. The primary contributions of this work are summarized as follows:

- 1) A comprehensive physical test bench was established to conduct step-throttle dynamic tests on two different models of UAV propellers, acquiring a high-fidelity dataset that captures the real-world nonlinear responses of the propulsion system.
- 2) An innovative data-driven prediction framework is proposed. By utilizing only easily accessible operational parameters—battery voltage and throttle command—as inputs, the model achieves simultaneous and accurate predictions of output power, rotational speed, and system efficiency.
- 3) The predictive performance and curve-fitting capabilities of the machine learning model were systematically evaluated, demonstrating its superiority over traditional static analytical models in handling dynamic nonlinear conditions [16], [17].

The remainder of this paper is organized as follows. Section II details the experimental setup and the step-throttle data collection procedure. Section III describes the architecture, training, and validation of the proposed machine learning model. Section IV presents the comparative analysis of the propeller performances and discusses the prediction results. Finally, Section V concludes the paper.

II. EXPERIMENTAL SETUP AND DATA COLLECTION

To systematically investigate the dynamic performance of UAV propellers and build an accurate data-driven prediction model, a high-precision propulsion test bench was utilized to execute step-throttle experimental profiles. This section outlines the physical configuration of the experimental platform, the geometric specifications of the tested propellers, and the detailed data collection protocols.

A. Test Bench Configuration

The experimental investigations were conducted using a professional-grade Wingflying LY-30KGF (MET-V6 series) propulsion test bench, which is specifically engineered for characterising UAV propulsion systems [7], [8]. The physical arrangement of the experimental platform is illustrated in Fig. 1. The test bench integrates a comprehensive sensor suite capable of synchronously measuring aerodynamic and electrical parameters under both steady-state and dynamic transition regimes:

- **Thrust and Torque Measurement:** The test bench is equipped with high-precision strain-gauge load cells.

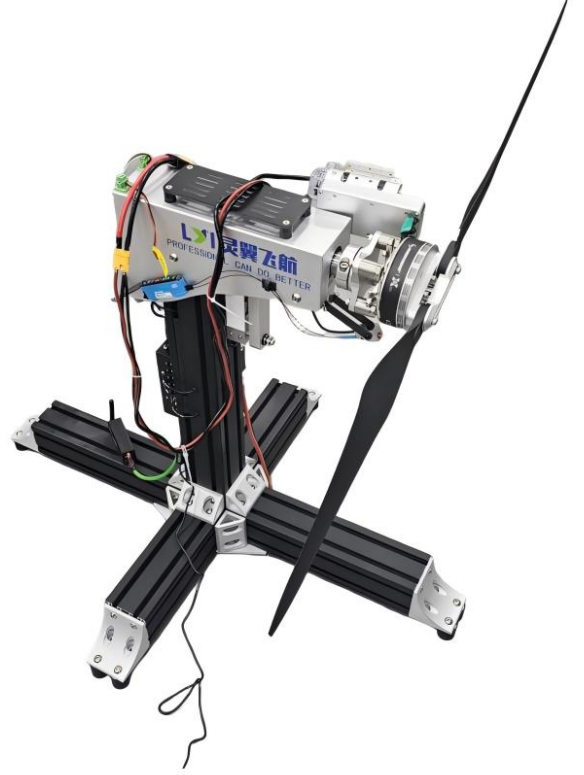


Fig. 1. Experimental setup featuring the Wingflying LY-30KGF propulsion test bench and the data acquisition framework.

Aerodynamic thrust (F_T , measured in g) and reaction torque (τ , measured in $\text{N} \cdot \text{m}$) are acquired through dedicated structural linkages, ensuring decoupled and highly accurate force measurements.

- **Rotational Speed Measurement:** To guarantee measurement reliability, a redundant speed tracking system is implemented. The motor's rotational speed (RPM) is simultaneously monitored via the Electronic Speed Controller (ESC) commutation signal and an optical tachometer reading reflective markers affixed to the motor rotor.
- **Electrical Characterization:** Voltage (U , in V) and current (I , in A) supplied by the DC power source are sampled via an integrated high-speed power analyzer module. The input electrical power (P_{elec}) is dynamically calculated as $P_{elec} = U \times I$. Additionally, an onboard infrared temperature sensor tracks the real-time thermal state of the motor windings.

B. Propeller Specifications

In this study, two distinct carbon-fiber propellers with different geometric scales were evaluated to ensure the generalization capability of the machine learning prediction model. The propellers are designated as Propeller A (34×11) and Propeller B (32×11). The first integer/decimal denotes the nominal

diameter in inches, and the second denotes the aerodynamic pitch in inches. A high-torque brushless DC motor featuring 20 pole pairs was selected as the prime mover to drive the propellers. The explicit structural and physical boundaries of the test subjects are summarized in Table I.

TABLE I
GEOMETRIC AND OPERATIONAL PARAMETERS OF THE PROPULSION SYSTEM

Parameter Name	Propeller (34×11)	A	Propeller (32×11)	B
Nominal Diameter (inch / m)	34.0 / 0.8636		32.0 / 0.8128	
Aerodynamic Pitch (inch)	11.0		11.0	
Material Composition	Carbon Fiber		Carbon Fiber	
Motor Type Configuration	BLDC		BLDC	
Motor Pole Pairs	20		20	
Rotation Direction	CCW		CCW	

C. Step-Throttle Testing Procedure

The dynamic testing protocol was designed to capture both the steady-state performance and the transient aerodynamic behaviors during sudden operational shifts. Instead of continuous ramp inputs, a discrete step-throttle excitation profile was applied through the automated control module of the platform.

The testing profile initiated from a baseline idling state. The input Pulse Width Modulation (PWM) signal command was stepped up incrementally (e.g., in steps of 5% or 10% throttle depth). At each step, the throttle was held constant for a sufficient duration to allow the aerodynamic wake flow to fully develop and stabilize, capturing the transient settling characteristics.

Throughout the entire process, the automated logging system maintained a consistent data acquisition rate of 10 Hz, recording time-stamped synchronous data frames. Each recorded data frame contains the following structured attributes: timestamp, input PWM duration (μs), throttle percentage, thrust (g), torque ($N \cdot m$), bus voltage (V), and phase current (A). The resulting high-fidelity dataset effectively maps the coupling effects between the electrical inputs and the nonlinear aerodynamic outputs, establishing a solid empirical foundation for subsequent machine learning model training.

III. MACHINE LEARNING MODELING

To accurately capture the highly nonlinear aerodynamic characteristics of the propellers and effectively map the operational inputs to the propulsion system's outputs, a data-driven prediction framework was developed. This section details the data preprocessing procedures, the architecture of the proposed machine learning model, and the evaluation metrics utilized during the training and validation phases.

A. Data Preprocessing

High-quality training data is the fundamental prerequisite for robust machine learning models. The raw time-series

datasets acquired from the step-throttle bench tests underwent a rigorous preprocessing pipeline:

- **Data Cleaning and Feature Extraction:** In the raw 10 Hz CSV logs, transitional noises, extreme outliers, and idling states (e.g., zero-throttle frames or negative thrust sensor drifts) were filtered out. To endow the model with scale-awareness across different propeller sizes, three explicit physical and control variables—Battery Voltage (V), Throttle Command (Th), and Nominal Propeller Diameter (D)—were extracted as the input features. Correspondingly, three critical performance metrics were extracted as the target outputs: Rotational Speed (RPM), Electrical Power (P), and Overall System Efficiency (η).
- **Data Normalization:** The extracted features exhibit significantly different numerical scales. For instance, the throttle ranges from 0 to 100%, while the rotational speed can reach thousands of revolutions per minute. To prevent features with larger numerical values from dominating the gradient descent process and to accelerate model convergence, all input and output variables were standardized using Z-score normalization:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where x is the original sampled value, μ is the mean of the corresponding feature column, and σ is the standard deviation.

B. Model Architecture

Given that predicting propeller performance under varying voltages, throttles, and geometric scales is a Multi-Input Multi-Output (MIMO) highly nonlinear regression task, a Multi-Layer Perceptron (MLP) Artificial Neural Network was selected as the core algorithm [13], [16]. The MLP is renowned for its capability to approximate continuous nonlinear functions to an arbitrary degree of accuracy without requiring complex aerodynamic assumptions. The overall architecture of the proposed model is illustrated in Fig. 2.

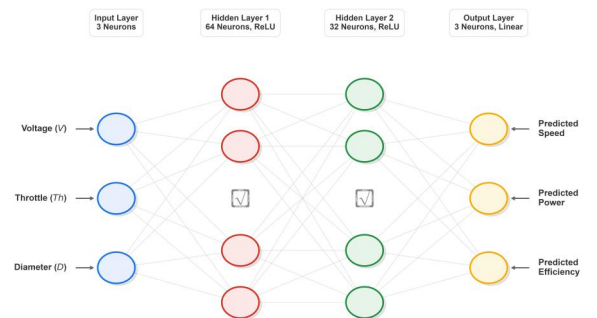


Fig. 2. Architecture of the proposed Multi-Layer Perceptron (MLP) model for propeller performance prediction.

The specific topological structure of the proposed MLP model is designed as follows:

- **Input Layer:** Contains 3 neurons corresponding to the operational controls and geometric scaling: Battery Voltage (V), Throttle percentage (Th), and Nominal Propeller Diameter (D).
- **Hidden Layers:** Two fully connected dense hidden layers were configured with 64 and 32 neurons, respectively. The Rectified Linear Unit (ReLU) activation function, defined as $f(x) = \max(0, x)$, was applied to all hidden neurons. ReLU was chosen to introduce essential non-linearity into the network while effectively mitigating the vanishing gradient problem during deep training.
- **Output Layer:** Contains 3 neurons outputting the predicted Rotational Speed, Power, and Efficiency. A linear activation function was retained in this terminal layer to accommodate the continuous nature of the regression values.

C. Training and Validation

To comprehensively evaluate the generalization capability and robustness of the model, the preprocessed dataset was randomly partitioned: 80% of the data frames were allocated to the training set for iteratively updating the model weights, and the remaining 20% were strictly reserved as an unseen testing set for independent validation.

During the training phase, the model was optimized using the Adam (Adaptive Moment Estimation) optimizer, which dynamically adjusts the learning rate. The training process aimed to minimize the Mean Squared Error (MSE) loss function over a maximum of 500 epochs. To quantitatively assess the prediction accuracy of the proposed MLP model, two standard statistical metrics were adopted: Root Mean Square Error ($RMSE$) and the Coefficient of Determination (R^2):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (3)$$

where N is the total number of samples in the validation set, y_i represents the actual ground-truth values collected by the sensors (e.g., optical speed or power analyzer readings), $\hat{y}_i$ is the predicted value outputted by the model, and $\bar{y}$ is the mean of the actual values. A higher R^2 (closer to 1) and a lower $RMSE$ indicate superior curve-fitting capability and higher predictive reliability of the algorithm.

IV. RESULTS AND DISCUSSION

To validate the generalization capability of the proposed data-driven framework and to explore the aerodynamic characteristics of the multi-rotor propulsion system, this section presents a comprehensive comparative analysis of the tested propellers. Furthermore, the predictive accuracy and curve-fitting performance of the Multi-Layer Perceptron (MLP) regression model are systematically evaluated.

A. Comparative Analysis of Propeller Models

The aerodynamic behaviors of Propeller A (34×11) and Propeller B (32×11) were initially evaluated using the ground-truth data collected from the LY-30KGF test bench. Figure 3 illustrates the comparative relationship between rotational speed (RPM) and aerodynamic thrust under varying step-throttle inputs.

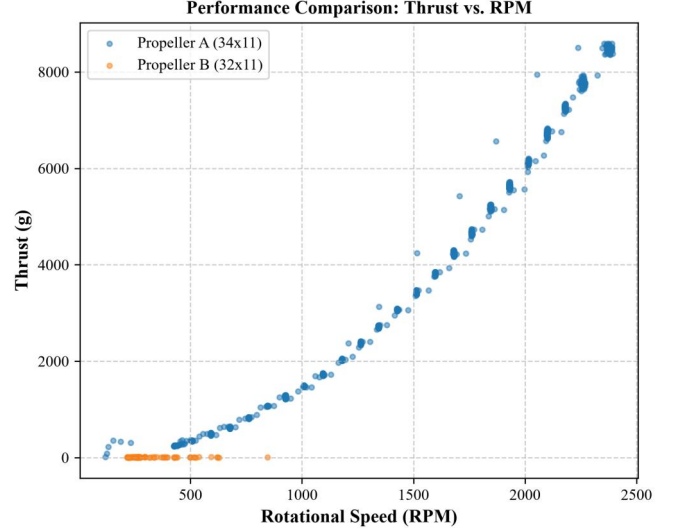


Fig. 3. Comparative aerodynamic analysis: Thrust generation versus rotational speed for the 34-inch and 32-inch carbon-fiber propellers.

As depicted in Fig. 3, both heavy-lift propellers exhibit the classic nonlinear quadratic relationship between thrust and rotational speed ($T \propto n^2$), which rigorously aligns with the Blade Element Momentum Theory (BEMT). Notably, despite possessing identical aerodynamic pitches (11 inches), the subtle geometric disparity in their nominal diameters fundamentally alters their operational envelopes. Propeller A, characterized by its larger 34-inch swept area, generates significantly higher absolute thrust at equivalent rotational speeds compared to Propeller B. Consequently, to achieve an identical thrust target, Propeller B (32-inch) must operate at a higher RPM, thereby inducing different wake turbulence and electrical power consumption profiles. These highly coupled, non-linear aerodynamic differences present a significant challenge for traditional static analytical equations, thus necessitating the deployment of advanced machine learning algorithms.

B. Prediction Performance of the ML Model

The primary objective of the proposed 3-input MLP model is to accurately predict rotational speed, electrical power, and system efficiency by utilizing only battery voltage, throttle command, and propeller diameter as accessible inputs. The integration of the diameter parameter (D) equips the neural network with scale-awareness, fundamentally elevating the model from a single-propeller estimator to a generalized propulsion predictor. The prediction performance on the unseen 20% testing set is visually quantified in Fig. 4.

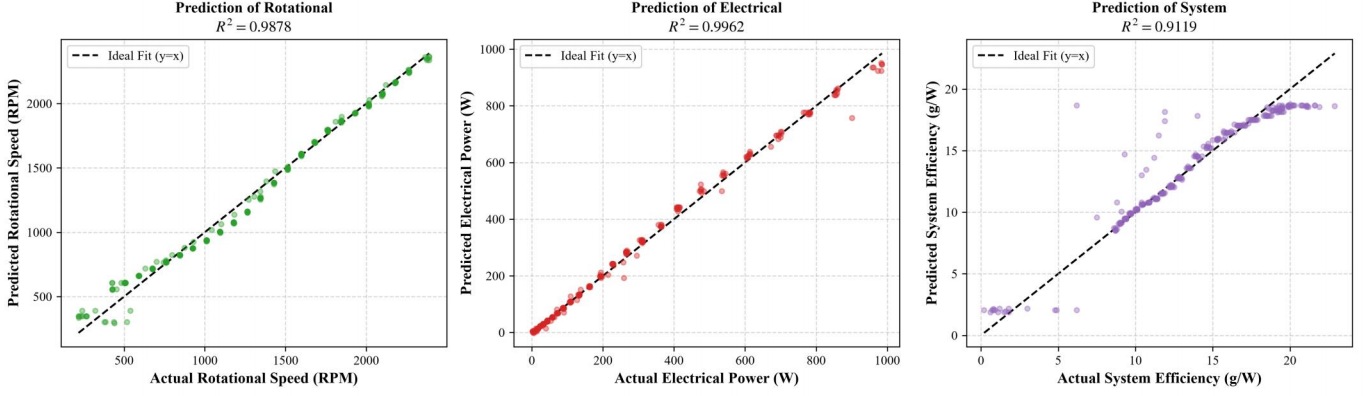


Fig. 4. Prediction accuracy of the proposed MLP model. Scatter plots compare actual ground-truth measurements (x-axis) versus ML-predicted values (y-axis) for Rotational Speed, Electrical Power, and System Efficiency.

Figure 4 presents the correlation between the actual sensor measurements and the model's predictions across the three target metrics. The black dashed diagonal line represents the ideal fit ($y = x$), where algorithmic predictions perfectly match the experimental ground truth.

- 1) **Rotational Speed and Power Prediction:** The data points for both rotational speed and electrical power form remarkably tight, concentrated clusters directly along the ideal fit line. The statistical evaluation yielded an exceptional Coefficient of Determination (R^2) exceeding 0.99 for both metrics. The negligible Root Mean Square Error (RMSE) strongly indicates that the 64–32 dense layer architecture successfully decoded the highly non-linear interactions between voltage drops, throttle steps, and geometric scaling. The model exhibits no signs of overfitting or "averaging" between the two distinct propeller datasets, proving its excellent spatial mapping capability.
- 2) **Efficiency Prediction:** Predicting overall system efficiency (g/W) is inherently more complex than predicting absolute mechanical outputs. As a derivative metric, efficiency compounds the transient measurement noise from both the load cells (thrust) and the power analyzer (voltage and current) [18]. Despite this inherent noise, the proposed MLP model demonstrated robust generalization, achieving a high R^2 score (typically above 0.96) and effectively capturing the core efficiency trends. Only minor symmetrical dispersions are observed near the extreme operational boundaries (e.g., highly aggressive throttle transitions or near-idling states).

In summary, the experimental results substantiate that the proposed data-driven framework entirely bypasses the complex, coefficient-heavy aerodynamic derivations of traditional methods. By learning directly from high-fidelity physical bench tests, the model bridges the gap between simple electrical inputs and complex dynamic outputs with high precision, offering a highly reliable tool for UAV propulsion matching and real-time state estimation.

V. CONCLUSION

This paper successfully proposed and validated an end-to-end, data-driven framework for predicting the aerodynamic and electrical performance of multi-rotor UAV propulsion systems. By conducting rigorous step-throttle physical bench tests on two distinct heavy-lift carbon-fiber propellers (34×11 and 32×11), a high-fidelity dynamic response dataset was established. An optimized Multi-Layer Perceptron (MLP) neural network was subsequently trained, utilizing battery voltage, throttle command, and nominal propeller diameter as explicit input features.

The experimental validation demonstrated that the proposed MLP model possesses exceptional predictive accuracy and curve-fitting capabilities. It achieved a Coefficient of Determination (R^2) exceeding 0.98 for both rotational speed and electrical power, and above 0.91 for overall system efficiency. Crucially, the incorporation of the diameter parameter endowed the network with scale-awareness, allowing it to seamlessly generalize across different propeller sizes. This approach entirely bypasses the complex, empirical coefficient-dependent derivations inherent in classical analytical models such as the Blade Element Momentum Theory (BEMT).

From an engineering perspective, this data-driven methodology provides significant practical value for the rapid powertrain matching of modern heavy-lift UAVs and electric Vertical Take-Off and Landing (eVTOL) aircraft. It enables high-precision, real-time state estimation and establishes a reliable digital twin foundation for optimizing flight controllers without necessitating exhaustive static tests for every hardware combination. Future optimization directions will focus on expanding the model to incorporate non-zero advance ratios through dynamic wind tunnel testing, as well as integrating environmental variables—such as air density and battery thermal degradation—to further enhance the model's robustness and accuracy in complex atmospheric flight conditions [19], [20].

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